Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2021	ENGI3491-WE01

Title:		
MEng/BEng: Digital Electronics & Digital Signal Processing		
Time Allowed:	2 hours	
Additional Material provided:	No	
Materials Permitted:		
Calculators Permitted:	Yes	Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: Yes		

Instructions to Candidates:	Answer ALL questions. All relevant workings must be shown.

Revision:	

Question 1

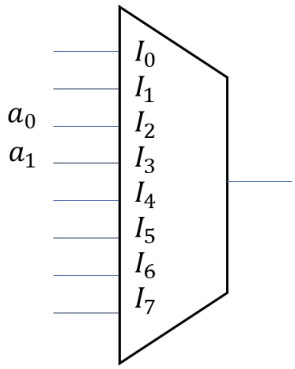
a) Implement the following function using a maximum of three 2-to-1 multiplexers:

$$F_1 = ab'g' + bcg' + df'g + fge$$

(where ‘ denotes NOT, e.g. b’ is NOT b)

[10%]

b) Implement a 4-to-1 multiplexer, using the 8-to-1 MUX shown below. The data inputs to the 4-to-1 MUX are a_0, a_1, a_2, and a_3, with a_3 being the most significant bit. a_0 and a_1 are already connected to the 8-to-1 MUX, as shown below. Name the control inputs of the 4-to-1 MUX as A and B.



[20%]

c) Using a PAL with 13 inputs and four outputs, implement an 8-to-1 multiplexer. Assume that each output OR gate of the PAL may have up to four AND inputs.

[30%]

d) Write a VHDL code for a signal generator with inputs T1 and T2 and an output, Out1. Both inputs T1 and T2 are 8-bit inputs, each representing an 8-bit integer. The output, Out1, stays at ‘0’ for T1 cycles and then at ‘1’ for T2 cycles, and this pattern is then repeated. Use a state machine for your design and draw its diagram. Also assume that in this signal generator, RESET is an active-high asynchronous input that resets the output Out1 to ‘0’ when asserted. Use standard logic.

[40%]

continued

Question 2

a) Compare and contrast asynchronous and synchronous sequential logic networks.

[30%]

b) Explain how and why flip flop metastability can cause digital system failure. Describe how such failure modes can be mitigated through the introduction of additional circuitry.

[20%]

c) An asynchronous sequential logic circuit is required to 'capture' two pulses from a clock-like signal, C, onto an external output, Z. A two-pulse sequence is defined as a rising edge -> falling edge -> rising edge -> falling edge transition sequence on C. The capture process is initiated when an external input, Y, changes from low to high. However, there are the following timing constraints:

- If Y transitions to high whilst C is already high then the two pulse capture will start at the next rising edge of C (provided Y is still high at that time).
- Two pulses from C will be captured even if Y transitions to low before both pulses have occurred on C.
- Once the two-pulse capture is complete, no further pulses can be captured regardless of any further changes to Y (or C).

i) Sketch an example timing diagram that is representative of this behaviour.

ii) Design an asynchronous sequential logic system (Moore type) to implement this specification. As well as full working, your solution should include all the simplified Boolean logic equations that are necessary to implement the circuit design.

continued

If you need to make any reasonable assumptions about the behaviour of the system in your design, please state them clearly.

iii) In general terms, briefly outline the approach that you would use to test that the circuit designed in (ii) performs as expected (n.b. you do not need to identify specific test inputs).

[50%]

END